

What is a Servlet?

A Servlet is a Java class that handles HTTP requests and generates responses in a web-based application.

It runs on a web server or application server like Apache Tomcat, Jetty, etc.

What is the lifecycle of a Servlet?

The lifecycle of a servlet is managed by the servlet container (e.g., Tomcat). It consists of the following

phases:

Instantiation: Servlet is instantiated once.

Initialization: `init()` method is called.

Request Handling: `service()` method is called for every client request.

Destruction: `destroy()` method is called before the servlet is unloaded.

What is the difference between `doGet()` and `doPost()` methods?

`doGet()`: Handles HTTP GET requests, typically used for fetching data from the server.

`doPost()`: Handles HTTP POST requests, generally used for submitting data (like form data) to the server.

What are the advantages of using Servlets?

Performance: Servlets are faster than CGI because they are loaded once and can handle multiple requests.

Platform Independence: Servlets are written in Java and can run on any platform that supports a servlet container.

Scalability: Servlets can efficiently handle a large number of concurrent requests.

What is the difference between a Servlet and JSP (Java Server Pages)?

Servlet: A servlet is a Java class that processes requests and generates responses. It is more code-intensive

and provides a programmatic approach to web development.

JSP: A JSP is a text-based page that is compiled into a servlet, and it is designed to simplify the presentation layer with HTML embedded in Java code.

What is the `web.xml` file in a web application?

The `web.xml` (deployment descriptor) file is used to configure servlet mappings, initialize parameters, security settings, and more in a web application.

What is a `ServletContext`?

The `ServletContext` is an interface that allows servlets to interact with the servlet container. It

provides information about the environment in which the servlet is running and is used for sharing data between servlets in the same web application.

What is the difference between `forward()` and `redirect()` in Servlets?

`forward()`: It forwards the request from one servlet to another servlet or JSP within the same server, without the client knowing.

`redirect()`: It sends a new request to a different URL and the client is aware of this change as the URL in the browser is updated.

What is the purpose of `init()` method in a servlet?

The `init()` method is called once when the servlet is loaded into memory by the servlet container. It is typically used for initial setup (like creating database connections or loading configuration data).

What is a session in Servlets?

A session is a mechanism to maintain state between client requests. The servlet container creates a session for each client, typically stored in a cookie (JSESSIONID) or URL rewriting.

How can you handle session management in a servlet?

Sessions can be managed using `HttpSession` in a servlet:

Creating a session: `HttpSession session = request.getSession();`

Storing data in session: `session.setAttribute("key", value);`

Retrieving data from session: `session.getAttribute("key");`

Invalidating a session: `session.invalidate();`

What is the difference between `doGet()` and `doPost()` in terms of security?

`doGet()`: Data sent through URL query parameters is visible in the browser's address bar, making it less secure for sensitive data.

`doPost()`: Data sent in the body of the HTTP request is not visible in the URL, making it more secure for transmitting sensitive information.

Explain the use of `ServletConfig` and `ServletContext` in servlets.

`ServletConfig`: Used to provide initialization parameters specific to a servlet.

`ServletContext`: Used to provide a global configuration for the entire web application.

What is the `@WebServlet` annotation in Servlets?

The `@WebServlet` annotation is used to declare a servlet without needing to configure it in `web.xml`. It simplifies servlet creation by providing an easier way to specify servlet attributes like `urlPatterns`.

What is the purpose of the `destroy()` method in a Servlet?

The `destroy()` method is called when the servlet is being unloaded from memory. It is used for cleanup tasks like closing database connections or releasing other resources.

What is the difference between `HttpServlet` and `GenericServlet`?

`HttpServlet`: Extends `GenericServlet` and is designed to handle HTTP requests (GET, POST, etc.). It provides convenience methods like `doGet()` and `doPost()`.

`GenericServlet`: A generic protocol-independent servlet that can handle any type of request and does not provide HTTP-specific methods.

What are some of the common performance optimization techniques for Servlets?

Use connection pooling for database connections.

Enable HTTP session clustering.

Implement caching mechanisms to reduce database load.

Avoid creating objects unnecessarily in the `doGet()` or `doPost()` methods.

Minimize I/O operations and optimize data retrieval.

How do you deploy a Servlet in a web application?

A servlet can be deployed by either:

Configuring it in the `web.xml` (deployment descriptor) file.

Using annotations (`@WebServlet`) to declare it directly in the servlet class.

What are the various HTTP status codes a Servlet might return?

200 OK: Successful request.

301 Moved Permanently: Resource has been permanently moved.

302 Found: Resource has been temporarily moved.

404 Not Found: Requested resource is not found.

500 Internal Server Error: Server encountered an error while processing the request.